WESP Data Pipeline

A reproducible pipeline for acquiring, cleaning, and structuring macroeconomic data for the United Nations World Economic Forecasting Model (WEFM)

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Overview

- replaces manual data downloads from websites with a reproducible automated workflow
- retrieves data via the SDMX API standard from international statistical providers (IMF, World Bank, Eurostat)
- implemented in R using Quarto documents, with extensive in-line documentation

Project Information:

- GitHub Repository (online Codesharing Platform): Link
- Maintainer: Marten W. - Global Economic Monitoring Branch (UN DESA)

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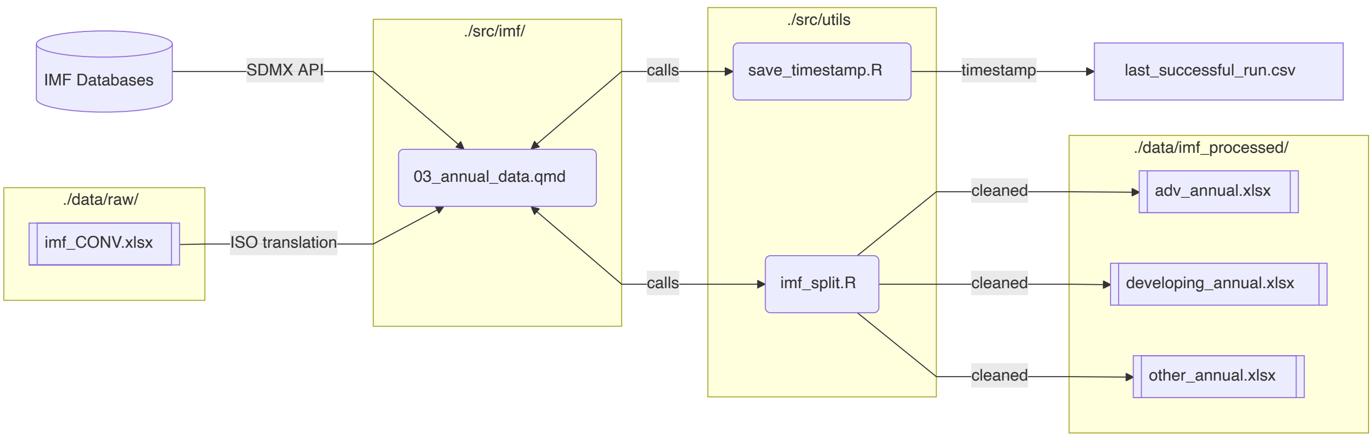
Workflow

The conceptual workflow of the pipeline is to retrieve datasets from external APIs, harmonize and transform them into WEFM-compatible structures, and export model-ready Excel files for ingestion into EViews.

Here is an example for IMF annual data:



1. access some IMF databases (e.g for Interest Rates or Balance of Payments) via the SDMX API

2. clean and transform the data using the *tidyverse* packages
3. add the relevant country codes for WEFM from `imf_CONV.xlsx`
4. split it into the country groupings using the utility script `imf_split.qmd`
5. save the excel files into the `data/imf_processed` folder
6. update the timestamp in `last_successful_run.csv`

Note: The final output data files are overwritten with each run. That's why there is a timestamp file, so you can see when they were last edited. If you open and edit the files in Excel, be sure to copy them to a different location; otherwise, your edits will be lost.

Step-by-Step Instructions

Requirements:

- R ≥ 4.2
- RStudio (recommended)
- Quarto
- Internet connection for API access

All package dependencies are managed through `renv`.

1. Go to the GitHub Code Repository (https://github.com/skriptum/WESP-Pipeline) and click on "Code > Download ZIP" to download the complete code
2. Unzip the Folder, move it to an appropriate location and open the `WESP_Pipeline.Rproj` file in RStudio
3. To recreate the environment, I used renv to keep track of the packages used. You need to restore the packages on my computer to make the project run on yours.

   ```
   install.packages("renv") # install renv if you don't have it yet
   renv::restore() # restore the environment
   ```

   - This could take some time, as downloading and extracting all packages takes some time
   - If it doesnt work, you need to go into each code file and download the packages that are required
4. Some workflows use the typical WEFM `Regions.xlsx` file. Replace it in `/data/raw` in case there are updates to country classifications / names etc.
5. Now, youre ready to run the code. The easiest way is to run the "Master" File `run-pipeline.qmd`. This file runs the process for all sources (IMF, WB, EUROSTAT), downloading and processing the data and places the finished files in the `./data/` folder
6. Verify that the pipeline executed successfully. Go into `last-successful-run.csv` and take a look at the table.
   - You should see the name of respective files, as well as their last update time and a "Success" in the status column
   - If that is not the case, go into the respective individual files to run them independently and see where they are failing

Feel free to go into each Quarto Code file and look at it before running it, there are a some more comments in there too to explain how it works

Project Structure

Below is a simplified overview of the repository structure:

```
└── data                          # data
```

```
├── data                    # data
│   ├── eurostat_processed  # processed (final) data sets
│   ├── imf_processed
│   ├── wb_processed
│   └── raw                 # raw data files used as inputs
├── docs
├── examples                # examples from the old WEFM
│   ├── eurostat
│   ├── imf
│   └── wb
├── renv                    # renv environment files
├── src                     # source code
│   ├── eurostat
│   │   ├── 01_examples.qmd          # ignore, just exploring the examples
│   │   ├── 02_national_accounts.qmd # annual and quarterly NA data
│   │   ├── 03_CPI.qmd               # Annual and monthly CPI
│   │   ├── 04_Unempl.qmd            # Annual and monthly Unemployment Rates
│   │   └── 05_File_combination.qmd  # File that builds the final xlsx file
│   ├── imf
│   │   ├── 01_example_exploration.qmd
│   │   ├── 02_monthly_data.qmd      # Exchange + Interest Rates
│   │   ├── 03_annual_data.qmd       # CPI, Current Account, ...
│   │   ├── 04_monthly_check.qmd     # checking overlap with old IFS
│   │   └── 05_statistics.qmd
│   ├── utils
│   │   ├── imf_split.R              # for IMF: split into advanced, developing
│   │   └── save_timestamp.R         # utility to timestamp the saving process
│   └── wb
│       ├── 01_wdi_download.qmd      # Poverty, PPP, Gini, ...
│       ├── 02_WEO_proxy_GGEI.qmd    # Interest rate proxy
│       └── 03_comparison.qmd        # of different Interest rate datasets
├── last_successful_run.csv # timestamp for last successful run of the pipeline
├── README.md               # this file
├── run-pipeline.qmd        # code file to run the whole pipeline at once
└── renv.lock               # renv lock file
```

Data Sources

IMF

Monthly Data:

IMF-Dataset	IMF Code	WEFM Code	WEFM Description
ER	XDC_USD; PA_RT	ENDA_XDC_USD_RATE / *_rfx_ncdol	Exchange rates, domestic currency per usd, period average, rate
MFS_IR	MFS166_RT_PT_A_PT	FPOLM_PA / *_rird	Financial, Interest Rates, Monetary Policy-Related Interest Rate, Percent per annum

Annual Data

IMF Dataset	IMF Code	WEFM Code	WEFM Description
ER	XDC_USD, PA_RT	ENDA_XDC_USD_RATE /*_rft_ncdol	Exchange rates, domestic currency per usd, period average, rate
CPI	CPI; _T; IX	PCPI_IX / *PCPI	Prices, Consumer Price Index, All items, Index
CPI	CPI; _T;YOY_PCH_PA_PT;	PCPI_PC_CP_A_PT / *PCPI_GR	Prices, Consumer Price Index, All items, Percentage change, Corresponding period previous year, Percent
WEO	BCA	BCAXF_BP6_USD / *BCANET$	Supplementary Items, Current Account, Net (Excluding Exceptional Financing), US Dollars
MFS_MA	NDMBM_MAI	FMB_XDC / *mnm2	National Definitions of Money, Base Money

WEO	GGXONLB - GGXCNL	GG_GEI_G01_XDC	Fiscal, General Government, Expense, Interest, 2001 Manual, Domestic Currency
QGFS	G24_T_XDC; S13	GG_GEI_G01_XDC_OLD / *gg_gei_g01_xdc	Fiscal, General Government, Expense, Interest, 2001 Manual, Domestic Currency (OLD)
QGFS	G24_T_XDC; S1311B	BCG_GEI_G01_XDC / *bcg_gei_g01_xdc	Fiscal, Budgetary Central Government, Expense, Interest, 2001 Manual, Domestic Currency

There are 2 different Indicators for General Government Interest Expense

- From the Quarterly Government Financial Statistics. This is the "official" IMF one, which is only available for ca 60 countries and a limited range of years

- one calculated from the WEO. It uses Government Balance data to calculate Interest Expense and lines up with the values from WB WDI, but with a larger coverage of countries
- both are still in the dataset, in case there is a decision to use the QGFS instead of WEO where available

The data sources used in the Calculation (from WEO):

- GGXONLB: Primary net lending (+) / net borrowing (-), General government, Domestic Currency
- GGXCNL: Net lending (+) / net borrowing (-), General government, Domestic Currency

Eurostat

The Links in the table already have the required preselection!

Indicators	**Unit**	**Adjustment**	**Eurostat Dataset ID**	**Code**
Annual GDP components	Chain Linked Volums (2010); Current Prices (millions of nat. curr)	None	NAMA_10_GDP	{ISO}_{CODE}_{ADJ}_ESTAT_Q
Quarterly GDP Components	Chain Linked Volums (2010); Current Prices (millions of nat. curr)	unadjusted; Seasonal; Seasonal + Calendar	NAMQ_10_GDP	{ISO}_{CODE}_{ADJ}_ESTAT_Q
Annual Unemployment	% of pop in labour force (age 15-74)	None	UNE_RT_A	{ISO}_URX_ESTAT
Monthly Unemployment	% of pop in labour force (total pop)	Seasonal	UNE_RT_M	{ISO}_URX_ESTAT_M
Annual HICP	All-items HICP, Annual Average	None	PRC_HICP_AIND	{ISO}_HIC_ESTAT
Monthly HICP	All-items HICP, 2015=100	None	PRC_HICP_MIDX	{ISO}_HIC_ESTAT_M

Annual GDP components are (Chain Linked Value code / Current Prices code)

- Gross domestic product at market prices (YER / YCN)
- Final consumption expenditure of general government (GCR / GCN)
- Household and NPISH final consumption expenditure (PCR / PCN)
- Gross fixed capital formation (ITR / ITN)
- Exports of goods and services (XTR / XTN)
- Imports of goods and services (MTR / MTN)

World Bank

Indicators taken from the World Bank WDIs

Description	WEFM Code	WDI Code
Interest payments (current LCU)	GGEI	GC.XPN.INTP.CN
GNI per capita, Atlas method (current US$)	GNICAP	NY.GNP.PCAP.CD
Gini index	GINI	SI.POV.GINI
Survey mean consumption or income per capita, total population (2021 PPP $ per day)	YBAR	SI.SPR.PCAP
Poverty headcount ratio at $3.00 a day (2021 PPP) (% of population)	HEAD	SI.POV.DDAY
PPP conversion factor, GDP (LCU per international $)	/	PA.NUS.PPP
GDP, PPP (current international $)	/	NY.GDP.MKTP.PP.CD
GDP, PPP (constant 2021 international $)	/	NY.GDP.MKTP.PP.KD